

# Hyperkalemia

*The three-phase approach: stabilize, shift, remove*

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Calcium stabilizes the myocardium — it does NOT lower the potassium.

# Why we get consulted

*The urgency comes from the arrhythmia risk, not the number.*

## TRIGGERS FOR CONSULT

- K  $\geq$  6.0 or any ECG changes
- Refractory despite medical therapy
- ESRD / CKD-5 needing KRT
- Hyperkalemia in AKI
- Recurrent on RAAS / MRA
- Binder selection (patiromer / SZC)

## TEACHING PEARL

*Three questions in order: is it real? Is the myocardium at risk (ECG)? What's the mechanism? Then treat. Skipping the ECG is the classic error.*

# First question: is this real?

*Rule out pseudohyperkalemia before you commit to aggressive therapy.*

## PSEUDOHYPERKALEMIA — consider if

- Hemolyzed sample (difficult draw, tourniquet)
- Fist clenching during phlebotomy
- Platelets > 400K or WBC > 50K (cells release K<sup>+</sup> in tube)
- Delayed transport (cold → K<sup>+</sup> leak)
- No ECG changes despite reported K<sup>+</sup> 7+

**Action: redraw (plasma, not serum if thrombocytosis); repeat ECG**

## TRUE HYPERKALEMIA — drivers

- ↓ Excretion: AKI, CKD 4–5, RAAS, MRA, K-sparing diuretics, heparin, calcineurin inhibitors, trimethoprim-sulfamethoxazole (TMP-SMX)
  - Cellular shift: acidosis, insulin deficiency (DKA), β-blockade, succinylcholine, digoxin toxicity
  - ↑ Load: tumor lysis, rhabdo, massive transfusion, K<sup>+</sup> supplements, salt substitutes
  - Hypoaldo: type 4 RTA (DM), adrenal insufficiency
- Action: treat the mechanism, not just the number**

# ECG: the driver of urgency

*ECG changes correlate poorly with  $K^+$  level — but any change means membrane stabilization now.*

| $K^+$ range      | Typical ECG                              | Action                                    |
|------------------|------------------------------------------|-------------------------------------------|
| <b>5.5 – 6.0</b> | Often none; occasional peaked T          | Hold RAAS/MRA; diet; consider binder      |
| <b>6.0 – 6.5</b> | Peaked T waves (narrow-based, tall)      | Shift therapy; binder; reassess every 2 h |
| <b>6.5 – 7.5</b> | PR prolongation, QRS widening, loss of P | Calcium NOW; shift; strongly consider KRT |
| <b>&gt; 7.5</b>  | Sine wave, VT/VF, asystole               | Calcium, shift, emergent KRT              |

**Classic teaching error:** waiting to treat until ECG changes appear. Some patients with  $K^+$  6.0 have widened QRS; others with  $K^+$  7 look normal. Treat based on rate of rise, AKI context, and ECG together — not  $K^+$  value alone.

# Three phases — don't mix them up

*Phase 1 protects the heart. Phase 2 buys time. Phase 3 removes the potassium.*

1

## STABILIZE

Protect the myocardium. Buys 30–60 minutes. Does NOT lower  $K^+$ .

2

## SHIFT

Move  $K^+$  intracellularly. Drops  $K^+$  0.5–1.0 mEq/L for 2–6 h. Does NOT remove  $K^+$ .

3

## REMOVE

Excrete  $K^+$  from the body. The only definitive step.

*Mixing phases is the classic trainee error: giving calcium and stopping, or giving insulin without a definitive removal plan.*

# Drugs, doses, and kinetics

Know onset and duration — it tells you when to re-check and re-dose.

| Drug                      | Dose                                           | Onset / Duration          | Notes                                                                                      |
|---------------------------|------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------|
| <b>Ca gluconate 10%</b>   | 10 mL IV over 2–5 min (can repeat ×1)          | 1–3 min / 30–60 min       | <i>STABILIZE only. Avoid peripheral Ca chloride (tissue necrosis). Caution on digoxin.</i> |
| <b>Insulin + dextrose</b> | 10 U reg IV (or 5 U if eGFR < 30) + 25 g D50   | 10–20 min / 4–6 h         | <i>Hypoglycemia in ~20–30% at 10 U in renal failure; glucose q1h × 6 h.</i>                |
| <b>Albuterol</b>          | 10–20 mg nebulized (4–8× the asthma dose)      | 30 min / 2–4 h            | <i>Additive with insulin. Avoid in active ischemia.</i>                                    |
| <b>NaHCO<sub>3</sub></b>  | 150 mEq in D5W over hours (if acidosis)        | Hours                     | <i>ONLY if metabolic acidosis — not a primary K<sup>+</sup> lowerer.</i>                   |
| <b>Furosemide</b>         | 40–80 mg IV if UOP present                     | 30–60 min / 4–6 h         | <i>Removes K<sup>+</sup> via urine; useless if anuric.</i>                                 |
| <b>Patiromer / SZC</b>    | Patiromer 8.4 g PO daily • SZC 10 g TID × 48 h | Patiromer ~7 h • SZC ~1 h | <i>GI binders. Outpatient control on RAAS. Not for life-threatening hyperK.</i>            |
| <b>Hemodialysis</b>       | K 2.0 or 1.0 bath × 3–4 h                      | Minutes / hours           | <i>Definitive. Removes 25–50 mEq/session. Expect rebound in 2–4 h.</i>                     |

# Urgent hyperkalemia algorithm

*From the moment you're paged: this is what happens in order.*

- 1** **Get the ECG immediately** Peaked T, wide QRS, or any change → emergency tier.
- 2** **Stabilize if ECG changes OR  $K \geq 6.5$**  Calcium gluconate 1 g IV. Re-dose if ECG not improved in 5 min.
- 3** **Shift** Insulin 10 U + D50 25 g IV. Albuterol 10–20 mg neb. Bicarb if pH < 7.20.
- 4** **Remove** Loop diuretic if UOP present. Binder (SZC/patiromer). Call dialysis if refractory or anuric.
- 5** **Fix mechanism, monitor** Hold RAAS/MRA/ $K^+$  supplements. Repeat  $K^+$  + ECG q1–2 h. Expect rebound in 2–4 h.

# Outpatient: keep the RAAS on

*Modern binders let us continue ACEi / ARB / finerenone / spironolactone in CKD.*

## BEFORE STOPPING RAAS

Re-check  $K^+$  — fasting, non-hemolyzed sample

Low- $K^+$  diet counseling (not salt substitutes!)

Diuretic if volume-tolerant (loop  $\pm$  thiazide)

Treat metabolic acidosis: oral  $NaHCO_3$  if  $HCO_3^- < 18$  (KDIGO 2024)

Stop K-sparing supplements, NSAIDs, bactrim

**Only after this — try a binder**

## BINDERS — PATIROMER vs SZC

Patiomer (Veltassa): 8.4–25.2 g PO daily. Takes 7 h for onset. Separate from other meds by 3 h. Constipation,  $\downarrow Mg^{2+}$ .

Sodium zirconium (Lokelma): 10 g TID  $\times$  48 h, then 10 g daily. Faster (1 h). High sodium load (caution HF).

**Both permit continuation of RAAS / MRA in CKD (AMBER trial — patiomer; DIAMOND — patiomer in HFrEF).**

Kayexalate (sodium polystyrene sulfonate (SPS)): avoid — risk of bowel necrosis; use SZC instead.

# Common pitfalls

*If you avoid these, you'll stay out of trouble.*

## AVOID

- Forgetting the ECG
- Assuming calcium lowered K
- Bicarb without acidosis
- Loop in anuria
- Kayexalate when SZC available
- Stopping RAAS permanently
- Missing insulin hypoglycemia

## DO INSTEAD

- Always present the ECG
- State: 'calcium stabilized; K still X'
- Check bicarb before  $\text{NaHCO}_3$
- Anuria → dialysis for removal
- SZC / patiromer, not Kayexalate
- Diet + binder before stopping RAAS
- Glucose q1h × 6 h after insulin

# Clinical case

*Think it through — answer is on the next slide.*

## VIGNETTE

A 68-year-old woman with T2DM, CKD 3b (eGFR 35), HFrEF 30%, on lisinopril 40 mg, spironolactone 25 mg, metoprolol, and furosemide 40 mg. ER check-in K<sup>+</sup> 6.8 after missing a week of follow-up. BP 138/80, HR 62. ECG: peaked T waves, PR 220, QRS 105 ms. Pt is asymptomatic.

**QUESTION:** What's the immediate sequence of interventions, and what's your outpatient plan?

# Clinical case — answer

*Key teaching points from the case.*

## ANSWER

**Calcium gluconate 1 g IV now, then shift (insulin 10 U + D50 25 g; albuterol 10 mg neb), then remove (loop diuretic if UOP, SZC 10 g TID).**

## TEACHING

Three phases in order: STABILIZE → SHIFT → REMOVE. Calcium protects the heart but does NOT lower K<sup>+</sup>. After shift, the K<sup>+</sup> rebounds in 4-6 h. Outpatient: her RAAS + MRA + CKD + diabetes are all driving hyperK, but these drugs are prognostically essential. Rather than stopping them permanently, add SZC or patiomer (AMBER, DIAMOND) and continue RAAS/MRA at adjusted doses. Diet counseling (avoid salt substitutes, bananas, potatoes) and adherence check are equally important.

## REFERENCES

# Hyperkalemia

### Guidelines

KDIGO 2024 CKD Guideline (potassium management)

ESC/AHA: hyperkalemia in HFrEF on MRA

ERA-EDTA position statement on hyperkalemia management

KDOQI commentary on dietary K<sup>+</sup> in CKD

### Landmark Trials

PEARL-HF — patiromer in HFrEF on MRA

OPAL-HK — patiromer for RAAS continuation

AMBER — patiromer in resistant HTN + CKD on spiro

DIAMOND — patiromer enables MRA in HFrEF

HARMONIZE — SZC maintenance of normokalemia

### UpToDate (2026 access):

*Causes and evaluation of hyperkalemia in adults • Clinical manifestations of hyperkalemia in adults • Treatment and prevention of hyperkalemia in adults*